Hope AI™
Your Future Starts Here

1. **Scenario:** A system checks if a user is eligible to vote based on their age.
   Write logic to ask the user for their age and determine if they are eligible to vote based on whether they are 18 or older.

   A:
   -First, we need to ask the age
   -Use age = int((input("Enter the age:")
   -And then use if age >= 18:
   -if the age of the person is created than or equal to 18 means print ("Eligible to vote")
   -use else statement, if the age is not created than or equal to 18, means it will print ("Not eligible to vote")

2. **Scenario:** A program processes a list of numbers and needs to find the largest value.
   Write logic to identify and return the largest number from a given list.

   A:
   -First ask a complete list of numbers.
   -For example, use list,
   -List= [4,45,52,48,23]
   -Use largest= max(List)
   -use print(largest)

3. **Scenario:** A company provides employees with a 10% bonus if their salary exceeds $50,000.
   Write logic to determine the bonus amount based on the given salary.

   A:

   -Ask the employee's salary details

   -Kindly put the int input statement

   -Kindly use if statement, if the salary statement is greater than $50,000, use 10% bonus.

   -if the salary is lesser than $50,000, set else statement, bonus to zero.

4. **Scenario:** A program evaluates a number to determine if it is even or odd.

   Write logic to check whether a given number is even or odd.

   A

   - Kindly add int input statement
   - Use if statement number %2 ==0:, print ("odd")
   - else statement, if the value is even, print ("even")

5. **Scenario:** A text-processing tool reverses a given word or sentence for formatting purposes.

   Write logic to take a word or sentence as input and produce its reversed version.

   A:

   -kindly assign word="hello world"
   -use reversed_text = text[::-1]
   -print(reversed_text)

6. **Scenario:** A grading system determines whether a student has passed or failed based on their score.

   Write logic to check if a student has passed a subject by scoring at least 40 marks.

   A:

   -Kindly add int input statement
   -Use if statement, value is greater than or equal to 40, print that you are cleared
   -use else statement, value is less than 40, print(you are not cleared)

7. **Scenario:** A retail store offers a 20% discount if a customer's total order exceeds $100. Write logic to calculate the final amount to be paid after applying the discount.

   A:

   -Kindly add int input statement
   -Use if statement, value is greater than or equal to 100, print that offer 20% discount
   -use else statement, value is less than 100, print does not provide 20% discount

8. **Scenario:** A banking system processes withdrawal requests and ensures the user has enough balance.

   Write logic to check if a user has enough balance before allowing a withdrawal and update the remaining balance accordingly.

   -Kindly add int input statement
   -Use if statement, value is equal to 0, print that you are insufficient

-use else statem ... process

9. **Scenario:** A calendar system verifies whether a given year is a leap year based on standard leap year rules.

   Write logic to determine whether a given year is a leap year.

   Add input value
   If the year is divisible by 400, it is a leap year.
   If the year is divisible by 100 but not by 400, it is not a leap year.
   If the year is divisible by 4 but not by 100, it is a leap year.
   Otherwise, it is not a leap year.

10. **Scenario:** A program filters out only even numbers from a given list.

   Write logic to extract and return only the even numbers from a list.

1. **Read the list of numbers.**
2. **Create an empty list to store even numbers.**
3. **Iterate through the list and check if each number is divisible by 2.**
4. **If divisible, add the number to the new list.**
5. **Return the list of even numbers.**